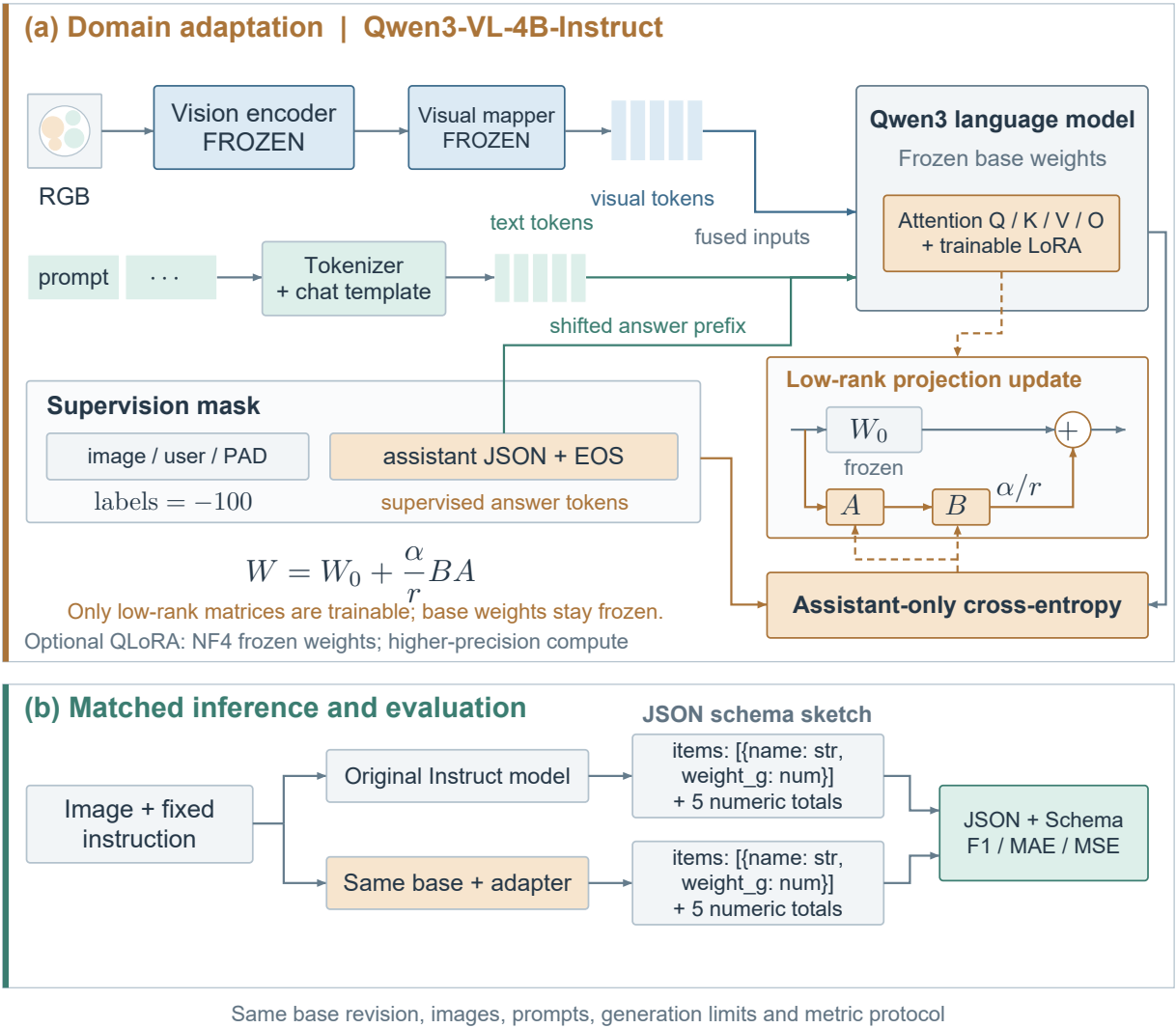

(a) Domain adaptation | Qwen3-VL-4B-Instruct
RGB
Vision encoder FROZEN
Visual mapper FROZEN
visual tokens
text tokens
fused inputs
Qwen3 language model
Frozen base weights
Attention Q / K / V / O + trainable LoRA
prompt
· · ·
Tokenizer + chat template
shifted answer prefix
Supervision mask
image / user / PAD
assistant JSON + EOS
labels = −100
supervised answer tokens
Low-rank projection update
$W_0$
frozen
$A$
$B$
$\alpha/r$
$W = W_0 + \frac{\alpha}{r} BA$
Only low-rank matrices are trainable; base weights stay frozen.
Optional QLoRA: NF4 frozen weights; higher-precision compute
Assistant-only cross-entropy
(b) Matched inference and evaluation
Image + fixed instruction
Original Instruct model
Same base + adapter
JSON schema sketch
items: [{name: str, weight_g: num}] + 5 numeric totals
items: [{name: str, weight_g: num}] + 5 numeric totals
JSON + Schema F1 / MAE / MSE
Same base revision, images, prompts, generation limits and metric protocol